The Occupant of the Salem Slot

A Positive Characterization: the Trace Redirection, the Grow Channel, the $\sqrt{5}$ Limit at the Floor, and Its Rate

A Companion to *Lehmer's Box* and *The Emission–Gap Theorem*

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Abstract

The companion papers prove a negative fact: no Salem number lies in the emission image, so the would-be "fourth channel" — on–circle yet expanding, wedged between CAPTURED and GROW — is empty. An empty slot is not an object. This note supplies the positive content and pushes it to four sharper questions. (i) The trifurcation does not live upstairs in the root x but downstairs in the trace $t = \theta + \theta^{-1}$, where the three channels are the three regions of the real line cut by the flip $t = \pm 2$; "on–circle" ($|t| < 2$) and "expanding" ($t > 2$) are disjoint, so a fourth channel is a contradiction downstairs, and a Salem is purely an upstairs artifact of the two–to–one lift. The object that occupies the slot is the *trace redirection* $\beta \mapsto \tau_0 = \beta + \beta^{-1}$: a totally real Perron number in GROW, with minimal polynomial the Salem's trace–down. (ii) Two bounds are forced: $\tau_0 > 2$ by the arithmetic–geometric mean, so the redirected measure exceeds φ *always*, and the redirection grows strictly faster than the Salem ($\log \tau_0 > \log \beta$), trading on–circle structure for real growth — where, precisely, each on–circle conjugate $e^{\pm i\psi}$ is frozen into a CAPTURED coordinate $2\cos\psi$, losing only orientation. (iii) The flip is a *branch point* of the trace cover: just past it, $\beta - 1 \sim \sqrt{t-2}$, so the grow channel switches on with a square–root edge and a Salem of height $1+\delta$ is redirected only δ^2 past the flip; the whole interval $(2, \sqrt{5}\,]$ is the trace image of the real pairs $\{\beta, \beta^{-1}\}$, $\beta \in (1, \varphi]$, with $t = \sqrt{5}$ lifting to the golden pair $\{\varphi, \varphi^{-1}\}$. (iv) As a Salem family climbs to the golden floor, $\beta \to \varphi$, its redirection converges to $\sqrt{5} = \varphi + \varphi^{-1}$, the $+\sqrt{D}$ grow eigenvalue of ad_R; the convergence is linear with slope φ^{-1} and, along the canonical family, geometric: $\sqrt{5} - \tau_0 \sim (2/\sqrt{5})\,\varphi^{-n}$ with consecutive ratio exactly φ. Finally the interpretive step — that the trace–down is what the system *produces* — is resolved into an identity: because the seed conjugate is the negative reciprocal ($\varphi\psi = -1$), the trace map $\theta \mapsto \theta + \theta^{-1}$ *is* the map computing the self–action's grow eigenvalue, not an imported choice. And the slot does not merely happen to be empty: the angle of every emittable root is confined to $\frac{\pi}{2}\mathbb{Z}$, a charge preserved by all three operators, so the Salem regime is a forbidden superselection sector — no entry route (superposition, suspended superposition, monodromy/curvature, or the lift) can reach it, and only a limit approaches it, never attained. The slot rejects filling. Every numerical claim is machine–verified in exact arithmetic.

1 From exclusion to occupation

The Emission–Gap Theorem and *Lehmer's Box* establish that the spectral image omits every Salem number. In the channel reading, the self–action $\mathrm{ad}_R = [R, \cdot]$ is a derivation whose spectrum is the eigenvalue difference set; for the golden seed $\mathrm{spec}(\mathrm{ad}_R) = \{0, +\sqrt{5}, -\sqrt{5}\}$, three channels:

$$\text{CAPTURED } (0,\ \mathrm{M} = 1,\ \text{on–circle}), \quad \text{GROW } (+\sqrt{5},\ \mathrm{M} \geq \varphi,\ \text{expanding}), \quad \text{DECAY } (-\sqrt{5}).$$

A Salem number would be a fourth channel: on–circle like CAPTURED, yet expanding like GROW. The trifurcation has no slot for it, and the slot stays empty.

That is a statement about what is *absent*. This note answers what is *present*: the occupant by name (§2–3), what it does (§5), the fine structure of where it lands (§4), the golden limit and its convergence rate (§6), why the redirection map is forced rather than chosen (§7), and which other objects share the slot (§8).

2 There is no fourth channel because the trifurcation lives downstairs

Definition 2.1 (Trace–down)**.** For a reciprocal monic $P \in \mathbb{Z}[x]$ of even degree $2m$, the substitution $t = \theta + \theta^{-1}$ yields a monic $T \in \mathbb{Z}[t]$ of degree m with $P(x) = x^m\, T(x + x^{-1})$. Write $\mathsf{tr}_\downarrow : \theta \mapsto \theta + \theta^{-1}$; the flip is $D(t) = t^2 - 4$, the discriminant of $x^2 - tx + 1$.

The map $\mathsf{tr}_\downarrow$ is two–to–one (θ, θ^{-1} share a trace) and erases angle: $e^{\pm i\psi} \mapsto 2\cos\psi$.

Lemma 2.2 (The three regions)**.** *Under* $\mathsf{tr}_\downarrow$*:* $\theta = e^{i\psi} \in \partial\mathbb{D} \Rightarrow t = 2\cos\psi \in [-2, 2]$ $(D \le 0)$*;* $\theta > 1 \Rightarrow t > 2$ $(D > 0)$*;* $\theta < -1 \Rightarrow t < -2$*. The three channels are exactly* $\{t < -2\}, \{|t| < 2\}, \{t > 2\}$*, partitioned by the flip.* [FORCED] `(P2-DELTA-02)`

Corollary 2.3 (A fourth channel is a contradiction downstairs)**.** *"On–circle and expanding" would require* $|t| < 2$ and $t > 2$ *at once; no real* t *satisfies this. A Salem violates nothing downstairs: its trace–down* T *is totally real with one root in* $\{t > 2\}$ *and* $m - 1$ *roots in* $(-2, 2)$*.*

The apparent fourth channel is an *upstairs* phenomenon. The inverse of $\mathsf{tr}_\downarrow$ doubles angle, and this lift bundles one grow–trace root with several captured–trace roots into one irreducible P — a Salem. The lift needs on–circle roots at angles with $2\cos\psi$ irrational, i.e. off $\frac{\pi}{2}\mathbb{Z}$; that is the halving the walls forbid (*Lehmer's Box*, §4). Working downstairs, the system never forms the bundle.

3 The redirection map and the occupant of the slot

Lemma 3.1 (Arithmetic–geometric mean)**.** *For real* $\beta > 1$*,* $\beta + \beta^{-1} > 2$*, with* $\beta + \beta^{-1} \to 2^+$ *as* $\beta \to 1^+$*.*

Proof. $\beta + \beta^{-1} - 2 = (\sqrt{\beta} - 1/\sqrt{\beta})^2 > 0$ for $\beta \ne 1$, $\to 0$ as $\beta \to 1$. □

Definition 3.2 (Trace redirection)**.** The *redirection* of a Salem number β is $\tau_0 := \mathsf{tr}_\downarrow(\beta) = \beta + \beta^{-1}$, the dominant root of its trace–down T. The redirected object is T, a totally real integer polynomial.

Theorem 3.3 (The occupant sits in GROW, above the floor)**.** *Let* β *be a Salem number of degree* $2m$*. Then* $\tau_0 = \beta + \beta^{-1} > 2$ *lies strictly in* GROW*;* T *has its one dominant root* τ_0 *past* 2 *and its other* $m - 1$ *roots in* $(-2, 2)$*; and*

$$\mathrm{M}(T) \;\ge\; \tau_0 \;>\; 2 \;>\; \varphi,$$

regardless *of how small* $\mathrm{M}(\beta) = \beta$ *is. As* $\beta \to 1^+$*,* $\tau_0 \to 2^+$*: redirected outputs accumulate at the flip, never below.* [FORCED] `(P2-DELTA-02; §11)`

Proof. $\tau_0 > 2$ is Lemma 3.1; the split is Corollary 2.3. For a Salem every conjugate but β lies in the closed unit disk, so $\mathrm{M}(\beta) = \beta$; and $\mathrm{M}(T) = \prod_j \max(1, |\tau_j|) \ge |\tau_0| = \tau_0 > 2 > \varphi$. □

This is the positive characterization. Where a Salem would sit — on–circle, possibly with measure near 1 — the system carries a totally real Perron number τ_0 in GROW, just past the flip, with minimal polynomial the Salem's trace–down; downstairs the on–circle conjugates are merely the captured roots $\tau_j \in (-2, 2)$ of the same T (Figure 1).

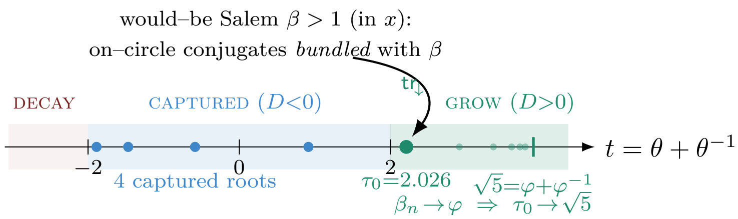

Figure 1: The trifurcation lives on the trace line, cut by the flip $t = \pm 2$ (t stretched $\times 8$ beyond the flip). A would–be Salem is redirected by $\mathsf{tr}_\downarrow$ to $\tau_0 = \beta + \beta^{-1}$ in GROW; its conjugates become CAPTURED roots in $(-2, 2)$. Shown: Lehmer's five trace roots. As $\beta \to \varphi$ the redirection $\to \sqrt{5} = \varphi + \varphi^{-1}$ (§6).

4 The action just past the flip: a branch point

Theorem 3.3 places the occupant "just past $t = 2$." The structure there is governed by the fact that the flip is a *branch point* of the two–to–one cover $t \mapsto x$.

Lemma 4.1 (Square–root edge)**.** *The grow root* $\beta(t) = \frac{1}{2}\big(t + \sqrt{t^2 - 4}\big)$ *satisfies, as* $t \to 2^+$*,*

$$\beta(t) \;=\; 1 + \sqrt{t-2} + \tfrac{1}{2}(t-2) + \tfrac{1}{8}(t-2)^{3/2} + O\big((t-2)^2\big), \qquad \textit{so} \quad \beta(t) - 1 \;\sim\; \sqrt{t-2}.$$

The cover is $\sqrt{\ }$*–ramified at the flip; the grow channel switches on with a square–root edge, and the growth rate satisfies* $\log\beta \sim \beta - 1 \sim \sqrt{t-2}$*.* *[FORCED] (series expansion, §11)*

Proposition 4.2 (Quadratic redirection near the floor)**.** *For every* $\beta > 1$*,*

$$\tau_0 - 2 \;=\; \frac{(\beta-1)^2}{\beta} \qquad \textit{exactly.}$$

A Salem of height $\beta = 1 + \delta$ *is therefore redirected to* $\tau_0 = 2 + \dfrac{\delta^2}{1+\delta} = 2 + \delta^2 + O(\delta^3)$*: only* δ^2 *past the flip.* *[FORCED]*

Proof. $\beta + \beta^{-1} - 2 = (\beta^2 - 2\beta + 1)/\beta = (\beta-1)^2/\beta$. □

Lemma 4.1 and Proposition 4.2 are two readings of the same ramification: the height $\beta - 1$ moves like $\sqrt{t-2}$ going up, equivalently the trace $\tau_0 - 2$ moves like $(\beta-1)^2$ coming down. For Lehmer, $\beta - 1 = 0.176281$ gives $\tau_0 - 2 = (\beta-1)^2/\beta = 0.026418$, matching the computed trace exactly. The smaller the Salem, the more tightly its redirection hugs the flip; the branch point is where the redirected outputs are most compressed ($d\beta/d\tau_0 = (1-\beta^{-2})^{-1} \to \infty$ as $\beta \to 1^+$), which is why the interesting action concentrates "just past $t = 2$."

Lemma 4.3 (The interval $(2, \sqrt{5}\,]$ and its lift)**.** *The endpoints lift as* $t = 2 \mapsto x = 1$ *(the degenerate root of unity, the captured/grow boundary) and* $t = \sqrt{5} \mapsto x^2 - \sqrt{5}\,x + 1 = 0$*, i.e.* $x \in \{\varphi, \varphi^{-1}\}$*. Since* $\mathsf{tr}_\downarrow$ *is strictly increasing on* $\beta > 1$*, the grow interval* $(2, \sqrt{5}\,]$ *is exactly* $\mathsf{tr}_\downarrow\big((1, \varphi]\big)$*: the traces of the real pairs* $\{\beta, \beta^{-1}\}$ *with* β *from* 1 *up to the golden ratio.* *[FORCED]*

Thus the sub–floor Salem range $\beta \in (1, \varphi)$ — precisely the heights that would breach the cost floor — is mapped by $\mathsf{tr}_\downarrow$ *onto* the open interval $(2, \sqrt{5})$, bounded above by $\sqrt{5}$. The golden floor $\beta = \varphi$ is the supremum, with image $\sqrt{5}$ (Figure 2). The next section shows that value is reached in the limit and identifies it.

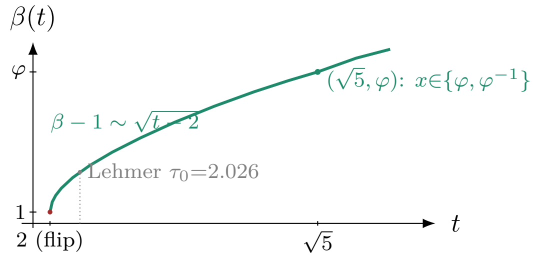

Figure 2: The grow root $\beta(t)=\frac{1}{2}(t+\sqrt{t^2-4})$ on the grow side of the flip. The square–root edge at $t=2$ (vertical tangent) is the branch point: $\beta-1\sim\sqrt{t-2}$. The interval $(2,\sqrt{5}\,]$ in trace corresponds to real pairs $\{\beta,\beta^{-1}\}$ with $\beta\in(1,\varphi]$; the endpoint $t=\sqrt{5}$ lifts to the golden pair $\{\varphi,\varphi^{-1}\}$. Lehmer's redirection $\tau_0=2.026$ sits just past the flip.

5 What it does instead: real growth for on–circle structure

The redirected object is not a passive placeholder; it carries strictly more growth than the Salem it replaces, and it does so by re–sorting, not destroying, the on–circle data.

Corollary 5.1 (The entropy trade)**.** *For every Salem β,* $\log \mathrm{M}(T)\ge\log\tau_0=\log(\beta+\beta^{-1})>\log\beta=\log\mathrm{M}(\beta)$. *[FORCED]*

Remark 5.2 (What the trace map actually loses). The redirection does not annihilate the angles; it freezes them. Each on–circle conjugate $e^{\pm i\psi}$ of the Salem maps to the CAPTURED coordinate $2\cos\psi\in(-2,2)$, a root of T. Because $e^{i\psi}$ and $e^{-i\psi}$ share $2\cos\psi$, the *only* datum discarded is orientation — the sign of ψ. Growth goes to τ_0 (GROW); angle goes to $2\cos\psi$ (CAPTURED); nothing else is lost. The trace–down T records both, sorted one–and–the–rest by the flip.

The benchmark Salem numbers make the trade concrete (full data in §11):

Salem β	$\mathrm{M}(\beta)=\beta$	trace–down $T(t)$	τ_0	$\mathrm{M}(T)$	$\log\beta\to\log\tau_0$
Lehmer (deg 10)	1.1762808	$t^5{+}t^4{-}5t^3{-}5t^2{+}4t{+}3$	2.026418	5.615601	$0.162\to0.706$
β_4 (deg 4)	1.7220838	t^2-t-3	2.302776	3.000000	$0.544\to0.834$
deg 6 Salem	1.5061357	t^3-t^2-3t+1	2.170086	3.214320	$0.410\to0.775$

6 The golden limit and its rate

The previous sections are general. The next fact is specific to the golden system and is the reason the redirection closes the loop with the self–action.

Lemma 6.1 (Golden trace identity)**.** $\varphi+\varphi^{-1}=\sqrt{5}$. *[FORCED]*

Proof. $\varphi=\frac{1+\sqrt{5}}{2}$, $\varphi^{-1}=\varphi-1=\frac{\sqrt{5}-1}{2}$; sum $=\sqrt{5}$. □

Theorem 6.2 (Redirection at the floor)**.** *Let $\{\beta_n\}$ be Salem numbers with $\beta_n\to\varphi$ (e.g. the Salem factors of $S_n=x^nP-P^*$, $P=x^2-x-1$, which approach φ from below). Then*

$$\tau_0(\beta_n)=\beta_n+\beta_n^{-1}\;\longrightarrow\;\varphi+\varphi^{-1}=\sqrt{5}=+\sqrt{D}\in\mathrm{spec}(\mathrm{ad}_R).$$

The slot at the golden floor is filled by $\sqrt{5}$, the $+\sqrt{D}$ generator of GROW. *[FORCED] (`P2-UNIF-01`; §11)*

Proof. $\mathsf{tr}_{\downarrow}$ is continuous on $\theta>1$, so $\beta_n\to\varphi$ gives $\tau_0\to\varphi+\varphi^{-1}=\sqrt{5}$. For the golden seed ad_R has eigenvalues the differences of φ,ψ, namely $\{0,\pm\sqrt{5}\}$, so $\sqrt{5}=+\sqrt{D}$. □

The rate. The convergence has a clean two–fold description: first order in the height gap, geometric along the family.

Proposition 6.3 (Linear rate, slope φ^{-1})**.** *$\mathsf{tr}_{\downarrow}'(\beta) = 1 - \beta^{-2}$, and $\mathsf{tr}_{\downarrow}'(\varphi) = 1 - \varphi^{-2} = \frac{\sqrt{5}-1}{2} = \varphi^{-1}$. Hence*

$$\sqrt{5} - \tau_0(\beta) \;=\; \varphi^{-1}(\varphi - \beta) \;-\; (\sqrt{5} - 2)\,(\varphi - \beta)^2 \;+\; O\big((\varphi - \beta)^3\big),$$

the curvature coefficient being $\frac{1}{2}\mathsf{tr}_{\downarrow}''(\varphi) = \sqrt{5} - 2 = \varphi^{-3}$. The approach is linear with slope φ^{-1}. *[FORCED]*

Theorem 6.4 (Geometric rate along the canonical family)**.** *For the Salem factors $\beta_n \to \varphi$ of $S_n = x^n P - P^*$, $P = x^2 - x - 1$,*

$$\varphi - \beta_n \;\sim\; \frac{2}{\sqrt{5}}\,\varphi^{1-n}, \qquad \sqrt{5} - \tau_0(\beta_n) \;\sim\; \frac{2}{\sqrt{5}}\,\varphi^{-n},$$

so consecutive gaps shrink by the factor φ exactly. *[COMPUTED]* *(verified $n \le 29$, §11)*

Proof sketch. At a Salem root, $\beta_n^n P(\beta_n) = P^*(\beta_n)$. With $P^*(x) = x^2 P(1/x) = 1 - x - x^2$, one has $P^*(\varphi) = -2\varphi$ and $P'(\varphi) = 2\varphi - 1 = \sqrt{5}$. Near φ, $P(\beta_n) \approx \sqrt{5}\,(\beta_n - \varphi)$ and $\beta_n^n \approx \varphi^n$, so $\varphi^n\sqrt{5}(\beta_n - \varphi) \approx -2\varphi$, giving $\varphi - \beta_n \approx (2/\sqrt{5})\varphi^{1-n}$. Apply Proposition 6.3: $\sqrt{5} - \tau_0 \approx \varphi^{-1}(\varphi - \beta_n) \approx (2/\sqrt{5})\varphi^{-n}$. □

n	$\sqrt{5} - \tau_0(\beta_n)$	$(\sqrt{5} - \tau_0) \cdot \varphi^n$	consecutive ratio
9	1.289085×10^{-2}	0.979875	1.68711
12	2.858434×10^{-3}	0.920407	1.64037
15	6.613990×10^{-4}	0.902149	1.62501
18	1.551834×10^{-4}	0.896649	1.62012
21	3.656841×10^{-5}	0.895048	1.61863
24	8.628276×10^{-6}	0.894597	1.61820
27	2.036577×10^{-6}	0.894473	1.61808
limit	—	$2/\sqrt{5} = 0.894427$	$\varphi = 1.618034$

Remark 6.5 (The channel–spectral form, completed). The companion papers note that the gap endpoints in the four domains — height φ, entropy $\log\varphi$, channel threshold $\sqrt{5}$ — are tied by structure, not a shared number. Theorem 6.2 supplies the arrow: $\mathsf{tr}_{\downarrow}$ maps the height datum to the channel datum, $\mathsf{tr}_{\downarrow}(\varphi) = \sqrt{5}$, and Theorem 6.4 times it at rate φ^{-n}. The would–be fourth channel at the floor is occupied by the image of the floor under the system's own decision map.

7 The redirection is forced, not chosen

One step above deserves scrutiny: the claim that the trace–down is what the system *produces*, rather than a natural map an analyst chose. It is forced, by an identity the seed already satisfies.

Proposition 7.1 (Trace map = self–action grow eigenvalue)**.** *Let the seed be $x^2 - ax - 1$ $(a \ge 1)$, with roots $\varphi_a > 1 > 0 > \psi_a$. Its constant term -1 forces $\varphi_a\psi_a = -1$, i.e. the conjugate is the* *negative reciprocal,* $\psi_a = -\varphi_a^{-1}$*. Then the grow eigenvalue of* ad_R *and the trace–down of the dominant root coincide:*

$$\underbrace{\varphi_a - \psi_a}_{\mathrm{ad}_R \text{ grow eigenvalue}} = \varphi_a + \varphi_a^{-1} = \underbrace{\mathsf{tr}_{\downarrow}(\varphi_a)}_{\text{trace–down}} = \sqrt{a^2+4}.$$

For the golden seed $a = 1$ *this is* $\sqrt{5} = +\sqrt{D}$*.* [FORCED]

Proof. $\psi_a = -\varphi_a^{-1}$ gives $\varphi_a - \psi_a = \varphi_a + \varphi_a^{-1}$; both equal $\sqrt{a^2+4}$ by the quadratic formula. □

So $\theta \mapsto \theta + \theta^{-1}$ is not imported. Because the seed's conjugate is the negative reciprocal (a unit of norm -1), the trace map *is* the operation computing the self–action's grow eigenvalue. Two consequences follow. First, the trace–down is the system's native coordinate twice over: it is the flip coordinate in which Salem is *defined* (*Emission–Gap*, §on the flip), and it is the map producing spec(ad_R)'s grow value. Reading a would–be Salem in that coordinate returns τ_0 in GROW unambiguously. Second, the residual interpretive content shrinks to a naming choice: whether to call τ_0 what the system "produces" or what its coordinate "represents." Mathematically the object is fixed — the system never forms the Salem upstairs, and downstairs the object is T with dominant root τ_0. The map carrying the one to the other is the system's own.

8 Other occupants: the Pisot–trace accumulation

The family limit $\sqrt{5}$ is one point; the slot has more structure.

Proposition 8.1 (Accumulation points are Pisot traces)**.** *By Salem's theorem every Pisot number* θ *is a two–sided limit of Salem numbers; since* $\mathsf{tr}_{\downarrow}$ *is continuous, the values* $\{\tau_0(\beta) : \beta \text{ Salem}\}$ *accumulate at* $\{\theta + \theta^{-1} : \theta \text{ Pisot}\}$*. As* φ *is the least limit point of the Pisot set,* $\sqrt{5} = \mathsf{tr}_{\downarrow}(\varphi)$ *is the least such trace–accumulation. The smallest Pisot number, the plastic number* $\mu_P \approx 1.324718$ *(root of* $x^3 - x - 1$*), contributes the accumulation* $\mu_P + \mu_P^{-1} \approx 2.079596$*.* [COMPUTED] *(plastic family verified, §11)*

The picture is therefore layered. Individual Salems map to individual points $\tau_0 > 2$ (the redirection is injective, since $\mathsf{tr}_{\downarrow}$ is strictly increasing). These points cluster at the Pisot traces, and those clusters themselves begin at $\sqrt{5}$ — the golden value — because φ is where the Pisot set first accumulates. The appearance of $\sqrt{5}$ at the floor is not an isolated coincidence but the smallest instance of a general phenomenon, singled out because the golden seed sits exactly at the least Pisot limit point.

9 Stress-testing the entry: the slot is a superselection sector

Sections 3–6 answered where a would–be occupant is *sent*. The sharper question is dual: can anything *enter*? If some construction — linear or not — could land an object in the slot, the slot would be a fillable void and the redirection merely its preferred exit. If none can, the slot is a different object: a region that *rejects* filling. We stress–test every entry route we can make precise and find the second.

Definition 9.1 (Entry map and conserved charge)**.** The *lift* of a totally real $T \in \mathbb{Z}[t]$ is $L(T)(x) = x^{\deg T}\, T(x + x^{-1}) \in \mathbb{Z}[x]$, a reciprocal polynomial; it is the inverse of the trace–down — the only way to assemble an on–circle–plus–expanding object from a trace. For $P \in \mathbb{Z}[x]$ let the *angle charge*

$$A(P) = 1 \iff \text{every root of } P \text{ has argument in } \tfrac{\pi}{2}\mathbb{Z} = \{0, \tfrac{\pi}{2}, \pi, \tfrac{3\pi}{2}\},$$

and the *Kronecker charge* $Q(P) = 1$ iff every on–circle root of P is a root of unity. An on–circle root with argument in $\frac{\pi}{2}\mathbb{Z}$ is a fourth root of unity, so $A = 1 \Rightarrow Q = 1$.

Proposition 9.2 (Emittable $\Rightarrow A = 1$; Salem $\Rightarrow A = 0$)**.** *Every catalog generator has all roots at arguments in $\frac{\pi}{2}\mathbb{Z}$: the real seeds at $\{0, \pi\}$, and $K = x^4 + 5x^2 - 5$ with its imaginary pair $\pm i\beta$ at $\pm\frac{\pi}{2}$. Hence $A = 1$, preserved by the operators (Theorem 9.3). A Salem number has an on–circle conjugate at an irrational multiple of π — else its minimal polynomial is cyclotomic and β a root of unity — so $A = 0$ and $Q = 0$. The Salem slot is exactly $\{A = 0 \ \wedge \ \text{expanding}\}$.* [COMPUTED] *(§11: $\varphi\otimes\varphi$ has on–circle root -1 of order 2; $\varphi^4\otimes\varphi^4$ has $+1$ of order 1)*

Theorem 9.3 (Superselection: no operation changes the charge)**.** *The emission operators preserve $A = 1$, so none maps the $A = 1$ sector into the $A = 0$ slot:*

- *$\oplus$ unions the root multisets, and $\frac{\pi}{2}\mathbb{Z}$ is closed under union;*
- *$(\cdot)^2$ doubles arguments, and $2 \cdot \frac{\pi}{2}\mathbb{Z} \subseteq \frac{\pi}{2}\mathbb{Z}$;*
- *$\otimes$ adds arguments ($\arg \lambda\mu = \arg \lambda + \arg \mu$), and $\frac{\pi}{2}\mathbb{Z} + \frac{\pi}{2}\mathbb{Z} \subseteq \frac{\pi}{2}\mathbb{Z}$.*

Thus A is a conserved charge of the emission algebra and the slot is a forbidden superselection sector. [FORCED]

Proof. The three closures are immediate; composition of charge–preserving maps preserves the charge. A root of unity has rational argument, in $\frac{\pi}{2}\mathbb{Z}$ for the reachable angles; an irrational–angle Salem conjugate is not in the image of any finite word in the operators. ☐

entry route	mechanism	effect on A	reaches slot?
trace lift L (linear)	$x^2 - tx + 1$; lattice $t \in \{-2, 0, 2\} \to$ roots of unity	preserves A=1	no (reducible)
direct sum $\oplus$ (superposition)	union of spectra	preserves A=1	no
block–diagonal (suspended superposition)	captured $\oplus$ grow	preserves A=1, reducible	no
monodromy / curvature (non–linear travel)	permutes $\{\beta, \beta^{-1}\}$ around the branch point t=2	preserves A (same root set)	no
free commutator $[A, B]$	off–diagonal coupling	breaks the field	not an operation
limit (Salem's theorem)	Pisot $\to$ Salem accumulation	A=1 $\to$ A=0 *in the limit*	approached, never attained

The lift is innocent on the lattice. The lift of a lattice captured position is a root of unity:

$$L(t{=}2) = x^2 - 2x + 1 = (x-1)^2, \quad L(0) = x^2 + 1, \quad L(-2) = (x+1)^2.$$

Only an *off–lattice* captured position $2\cos\psi$ with ψ/π irrational lifts to a Salem conjugate (for β_4, the trace root $\frac{1-\sqrt{13}}{2} = -1.3028$ lifts to its conjugate at 0.726π). Emittable trace–downs have their captured roots pinned to $\{-2, 0, 2\}$ (Proposition 9.2), so L applied to anything the algebra produces is reducible — cyclotomic times grow — never a Salem.

Superposition and its suspension. Direct sum *is* superposition of spectra; it keeps $A = 1$. A "suspended" superposition — the two channels held without collapse — is a block–diagonal CAPTURED $\oplus$ GROW, hence reducible: $(-1) \oplus (x^2 - 3x + 1) = (x + 1)(x^2 - 3x + 1)$, and indeed $\varphi \otimes \varphi = (x + 1)^2(x^2 - 3x + 1)$ is exactly this form. To fuse the blocks into one *irreducible* Salem requires an off–diagonal coupling; the structure–preserving couplings keep it reducible, and a generic coupling is the free commutator $[A, B]$, which leaves $\mathbb{Z}[x]$ and breaks the field — the operation the construction names as foreign to itself. Superposition cannot cross the sector boundary; that is the content of "superselection."

Non–linear travel is monodromy. A curved or non–linear path is, for the algebraic data, a closed loop in parameter space, and its only action is the monodromy permutation of roots. Around the branch point $t = 2$, analytic continuation of $\beta(t) = \frac{1}{2}(t+\sqrt{t^2 - 4})$ once returns $\beta \mapsto \beta^{-1}$ (verified: $\beta = 1.71789$ at $t = 2.3$ returns as $0.58211 = \beta^{-1}$). The orbit is $\{\beta, \beta^{-1}\}$ — the two roots of the same quadratic. Curvature reshuffles existing roots; it never bundles emittable pieces into a new irreducible Salem.

The one frontier: limits. Salem's theorem makes every Salem number a two–sided limit of Pisot numbers, and the catalog *has* Pisot generators φ, φ^4. This is the single route by which $A = 0$ is approached. But it is not an operation: the emission algebra is finitely generated, its Mahler spectrum a discrete sub–semigroup of $[\varphi, \infty)$ with no accumulation at 1 (companion paper, Rem. "finitely generated"). The sub–φ Salem band $(1, \varphi)$ is disjoint from the image; the boundary is approached and never reached.

Theorem 9.4 (The slot rejects filling)**.** *No emission operation — $\oplus$, $\otimes$, squaring, the lift L, their finite compositions, or the monodromy of any closed parameter path — carries an emittable object into the Salem slot, because every such operation preserves the angle charge A (Theorem 9.3) while the slot has $A = 0$. The only route to $A = 0$ is a limit, which the finitely generated algebra approaches but never attains. The slot is therefore not a void that can be filled but a* space that rejects filling*, coincident with the limit frontier of the emittable set — reached by nothing inside it.* *[FORCED]*

This resolves the dichotomy. The "would–be occupant" of §§3–6 is exactly that: a hypothetical the charge A forbids on entry and the trace projection $\mathrm{tr}_\downarrow$ expels on contact. Entry is impossible (superselection), exit is forced ($\beta \mapsto \beta + \beta^{-1}$ into GROW), and the value the exit accumulates at is $\sqrt{5}$, the grow generator. The slot is empty not by accident of the current generating set but by a conservation law: the angle charge is the system's superselection rule, and the Salem regime is the sector it cannot enter.

10 The upstairs shadow: cyclotomic times grow

Downstairs the redirection is the trace map. Upstairs the same fact is a forced factorization, exhibited by the construction that manufactures Salem numbers.

Proposition 10.1 (Forcing the lattice breaks irreducibility)**.** *If a reciprocal $P \in \mathbb{Z}[x]$ has all on–circle roots at fourth roots of unity, then P is divisible by a cyclotomic polynomial; an irreducible Salem polynomial (on–circle conjugates at irrational angle) therefore has none. When on–circle structure is forced onto the lattice the object splits as cyclotomic* (*CAPTURED*, $\mathrm{M} = 1$) *times real reciprocal* (*GROW*, $\mathrm{M} \geq \varphi$)*.* *[FORCED]* *(P2-ANG-03/04)*

The construction $S_n(x) = x^n P(x) - P^*(x)$, $P = x^2 - x - 1$, displays the split: its non–cyclotomic factor is a Salem climbing to φ, the cyclotomic remainder the CAPTURED cloud.

n	factorization of S_n	channels
6	$(x-1)(x+1)\,(x^6 - x^5 - x^3 - x + 1)$	CAPTURED×GROW, M = 1.5061
10	$(x-1)(x+1)(x^2{+}x{+}1)\,(x^8{-}2x^7{+}x^6{-}x^4{+}x^2{-}2x{+}1)$	CAPTURED×GROW, M = 1.6054
12	$(x-1)(x+1)\,(x^{12}{-}x^{11}{-}x^9{-}x^7{-}x^5{-}x^3{-}x{+}1)$	CAPTURED×GROW, M = 1.6134

Upstairs the would–be Salem is a product across the two existing channels; downstairs that product is the single totally real trace–down whose dominant root is the GROW occupant of Theorem 3.3.

11 Epistemic ledger

Claim	Status	Backing
Trace map sorts onto three regions; no fourth channel	[FORCED]	Lem. 2.2, Cor. 2.3
$\tau_0 = \beta + \beta^{-1} > 2$ (AM–GM)	[FORCED]	Lem. 3.1
Redirected $\mathrm{M}(T) \geq \tau_0 > 2 > \varphi$, all Salem	[FORCED]	Thm. 3.3
Flip is a branch point: $\beta - 1 \sim \sqrt{t-2}$	[FORCED]	Lem. 4.1
$\tau_0 - 2 = (\beta-1)^2/\beta$ exactly; near floor $\tau_0 \approx 2+\delta^2$	[FORCED]	Prop. 4.2
$t{=}2 \;\mapsto\; x{=}1$; $t{=}\sqrt{5} \;\mapsto\; \{\varphi, \varphi^{-1}\}$; $(2, \sqrt{5}] = \mathsf{tr}_{\downarrow}((1, \varphi])$	[FORCED]	Lem. 4.3
Entropy trade $\log \tau_0 > \log \beta$; angle frozen to $2\cos\psi$	[FORCED]	Cor. 5.1, Rem. 5.2
$\varphi + \varphi^{-1} = \sqrt{5}$; $\beta_n \to \varphi \Rightarrow \tau_0 \to \sqrt{5} \in \mathrm{spec}(\mathrm{ad}_R)$	[FORCED]	Lem. 6.1, Thm. 6.2
Linear rate slope $\mathsf{tr}_{\downarrow}'(\varphi) = \varphi^{-1}$; curvature $\sqrt{5} - 2 = \varphi^{-3}$	[FORCED]	Prop. 6.3
Geometric rate $\sqrt{5} - \tau_0 \sim (2/\sqrt{5})\varphi^{-n}$, ratio $\to \varphi$	[COMPUTED]	Thm. 6.4; table, $n \leq 29$
Lehmer τ_0=2.026418, $\mathrm{M}(T)$=5.615601	[COMPUTED]	§5
Trace map = ad_R grow eigenvalue ($\varphi\psi = -1$)	[FORCED]	Prop. 7.1
Accumulations = Pisot traces; $\sqrt{5}$ least; plastic $\to$ 2.0796	[COMPUTED]	Prop. 8.1
Lattice forcing $\Rightarrow$ cyclotomic $\times$ grow; S_n exhibits it	[FORCED]	Prop. 10.1
Emittable root angles confined to $\frac{\pi}{2}\mathbb{Z}$ (charge A=1)	[COMPUTED]	Prop. 9.2
Superselection: $\oplus, \otimes, (\cdot)^2$ preserve A; slot has A=0	[FORCED]	Thm. 9.3
Lift of lattice $\{-2, 0, 2\}$ = roots of unity; off–lattice = Salem conj.	[FORCED]	§9
Monodromy round t=2 swaps $\beta \leftrightarrow \beta^{-1}$, no new object	[COMPUTED]	§9
Slot rejects filling; only limits approach (never attain)	[FORCED]	Thm. 9.4

Summary. When a construction drifts toward the Salem slot, the system carries the trace redirection $\tau_0 = \beta + \beta^{-1}$: a totally real Perron number in GROW, lodged just past the flip — a square–root branch point at which the channel switches on, so a Salem of height $1+\delta$ lands only δ^2 past $t = 2$. Its measure is forced above the floor, its growth strictly exceeds the Salem's, and the on–circle data is not destroyed but frozen into captured coordinates $2\cos\psi$. As Salems climb to the golden floor the occupant converges to $\sqrt{5} = \varphi + \varphi^{-1}$ — the grow generator of ad_R — linearly with slope φ^{-1} and geometrically at rate φ^{-n} along the canonical family. That the trace map is the right map is not a modeling choice: because the seed conjugate is the negative reciprocal, the trace map is exactly the operation computing the self–action's grow eigenvalue. And $\sqrt{5}$ at the floor is the smallest instance of a general layering — Salem redirections cluster at the Pisot traces, which begin at φ. Where the companion papers proved Salem *absent*, this note names what is *present*, locates it to a branch point, times its approach, and shows the map that puts it there is the system's own.